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CHEMISTRY

1) At 35°C, the vapour pressure of CS_2 is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is :

(1) heat must be absorbed in order to produce the solution at 35°C

(2) Raoult's law is not obeyed by this system

(3) a mixture of 100 mL CS_2 and 100 mL acetone has a volume < 200 mL

(4) CS_2 and acetone are less attracted to each other than to themselves

2) Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non volatile solute are together sealed in a container. Over time -

(1) The volume of the solution does not change and the volume of the solvent decreases

(2) The volume of the solution decrease and the volume of the solvent increases

(3) The volume of the solution increase and the volume of the solvent decreases

(4) The volume of the solution and the solvent does not change

3) The freezing point of a diluted milk sample is found to be –0.2°C, while it should have been –0.5°C for pure milk. How much water has been added to pure milk to make the diluted sample ?

(1) 2 cups of water to 3 cups of pure milk

(2) 1 cup of water to 3 cups of pure milk

(3) 3 cups of water to 2 cups of pure milk

(4) 1 cup of water to 2 cups of pure milk

4) What is the mole ratio of benzene (P_B^0= 150 torr) and toluene (P_T^0 = 50 torr) in vapour phase if the given solution has a vapour pressure of 120 torr ?

(1) 7 : 1

(2) 7 : 3

(3) 8 : 1

(4) 7 : 8

5) Correct order of freezing point of given aqueous solution :
Solution-I : 0.1m KCl,
Solution-II : 0.2 m HA (α = 0.5)
Solution-III : 0.1m $C_6H_{12}O_6$

(1) I < II < III

(2) III < II < I

(3) III < I < II

(4) II < I < III

6) In a binary ideal liquid solution of A and B in which vapour pressure of pure component A and B are 600 and 400 torr respectively at temperature 'T'. Pick the correct statement :

(1) mole fraction of A in vapour phase is always greater than mole fraction of B in vapour phase.

(2) mole fraction of A in vapour phase is always greater than mole fraction of A in liquid phase.

(3) vapour pressure of solution can not exceed 600 torr at temperature T.

(4) mole fraction of B in vapour phase is always greater than mole fraction of B in liquid phase.

7) A solution of x moles of sucrose in 100 grams of water freezes at –0.2ºC. The amount of ice that will separate out when it is cooled upto –0.25°C is :-

(1) 18 grams

(2) 20 grams

(3) 25 grams

(4) 23 grams

8)

At 27°C, a 0.1 M aqueous solution of a weak electrolyte A_2B has an osmotic pressure of 6.15 atm. The degree of dissociation α of the electrolyte is: (Use $R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}$)

(1) 0.50

(2) 0.75

(3) 1.00

(4) 0.25

9) Insulin is dissolved in a suitable solvent & the osmotic pressure (π in atm) of solutions of various concentration **'C' in gm/cc** is measured at 27°C. The slope of the plot π against **'C'** is found to be 5×10^{-3} . The molecular weight of insulin is : [R = 0.0821 litre-atm/mol-K]

(1) 4.9×10^5

(2) 4.9×10^3

(3) 4.9×10^6

(4) 4.9×10^4

10) Acetic acid (CH_3COOH) exists in the dimeric form in a solution. In an actual experiment, the Van't Hoff factor was found to be 0.52. The degree of association and osmotic pressure of 0.1 M acetic acid solution at 300K respectively are : (R = 0.0821 litre-atm/K-mole)

(1) 0.96 and 9.6 atm

(2) 0.96 and 1.28 atm

(3) 0.46 and 1.28 atm

(4) 0.96 and 7.86 atm

11) At a given temperature, total vapour pressure in Torr of a mixture of volatile components A and B is given by
$P_{Total} = 120 - 75\ X_B$
hence, vapour pressure of pure A and B respectively (in Torr) are

(1) 120, 75

(2) 120, 195

(3) 120, 45

(4) 75, 45

12) Among the following, that does not form an ideal solution is :

(1) C_6H_6 and $C_6H_5CH_3$

(2) C_2H_5Cl and C_6H_5OH

(3) C_6H_5Cl and C_6H_5Br

(4) C_2H_5Br and C_2H_5I

13) A solution of a substance containing 1.05 g per 100 mL. was found to be isotonic with 3% glucose solution. The molecular mass of the substance is -

(1) 31.5

(2) 6.3

(3) 630

(4) 63

14) Which has maximum osmotic pressure at temperature T :

(1) 100 mL of 1 M urea solution

(2) 300 mL of 1 M glucose solution

(3) mixture of 100 mL of 1 M urea solution and 300 mL of 1 M glucose solution

(4) all are isotonic

15) Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mmHg, respectively, at the same temperature. Then correct statement is:
(x_M = Mole fraction of 'M' in solution ;
x_N = Mole fraction of 'N' in solution ;
y_M = Mole fraction of 'M' in vapour phase ;
y_N = Mole fraction of 'N' in vapour phase)

(1) $(x_M - y_M) < (x_N - y_N)$

(2) $\frac{x_M}{x_N} < \frac{y_M}{y_N}$

(3) $\frac{x_M}{x_N} > \frac{y_M}{y_N}$

(4) $\frac{x_M}{x_N} = \frac{y_M}{y_N}$

16) The value of observed and calculated molecular weight of silver nitrate are 92.64 and 170 respectively. The degree of dissociation of silver nitrate is :

(1) 60%

(2) 83.5 %

(3) 46.7%

(4) 60.23%

17) If Ar gas is bubbled through water at 298 K, how many moles of Ar gas would dissolve in 1 litre of water? Assume that Ar exerts a partial pressure of 0.403 bar. Given that Henry's law constant for Ar at 298 K is 40.3 K bar

(1) 6.02×10^{-3}

(2) 1×10^{-5}

(3) 18×10^{-5}

(4) 55.56×10^{-5}

18) Glucose is added to 1 litre water to such an extent that $\frac{\Delta T_f}{K_f}$ becomes equal to $\frac{1}{1000}$ the wt. of glucose added is

(1) 180 g

(2) 18 g

(3) 1.8 g

(4) 0.18 g

19)

Pure water boils at 373 K and nitric acid at 359 K. The azeotropic mixture of water and nitric acid boils at 393.5 K. On distillation of the azeotropic mixture.

(1) pure nitric acid will distil over first.

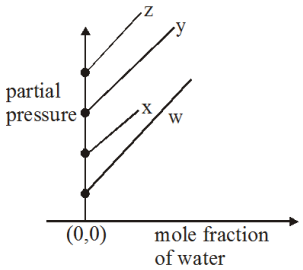
(2) pure water will distil over first.

(3) one of them will distil over with small amount of the other.

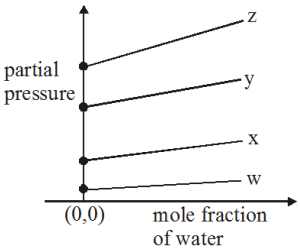
(4) both of them will distil over in the same composition as they are in the mixture.

20) For the solution of the gases w, x, y and z in water at 298K, the Henrys law constants (K_H) are 0.5, 2, 35 and 40 kbar, respectively. The correct plot for the given data is :-

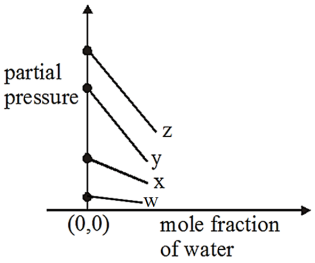
(1)



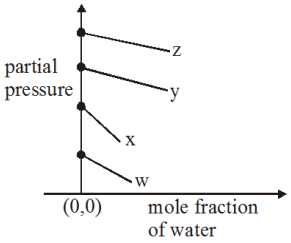
(2)



(3)



(4)



21) Which of the following statement is **incorrect** ?

(1) The tanks used by Scuba driver are filled with N_2 and O_2

(2) Swelling of tissue cells is called edema.

(3) Osmotic pressure is use for determination of molecular mass of biomolecules which are not stable at high temperature

(4) Mixture contain 68% HNO_3 & 32% H_2O by mass is example of maximum boiling azeotrope.

22) Boiling point of a 2% aqueous solution of a non-volatile solute A is equal to the boiling point of 8% aqueous solution of a non-volatile solute B. The relation between molecular weights of A and B is.

(1) $M_A = 4M_B$

(2) $M_B = 4M_A$

(3) $M_A = 8M_B$

(4) $M_B = 8M_A$

23) Given below are two statements :
Statement-I : A mixture of phenol and aniline will show positive deviation from Raoults law .
Statement-II : When intermolecular intractions weaken after mixing liquids then vapour pressure increases.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement-I is false but Statement-II is true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is true but Statement-II is false

(4) Both Statement-I and Statement-II are true

24) Match List-I (solutes) with List-II (ratio of depression in freezing point) and select the correct answer from the codes given below :

List-I [Solute 0.01 m]		**List-II** [ratio of depression in freezing point]	
(A)	Urea, NaCl, Na_2SO_4	(a)	1:2:1
(B)	KCl, NaCl, K_2SO_4	(b)	1:2:3
(C)	Urea,Sugar,Glucose	(c)	2:2:3
(D)	Urea, KCl, Glucose	(d)	1:1:1

Code is :-

	A	**B**	**C**	**D**
(1)	b	c	d	a
(2)	a	b	c	d
(3)	b	a	c	d
(4)	c	d	b	a

(1) 1

(2) 2

(3) 3

(4) 4

25) Match the solute and solvent in column-I with examples in column-II and mark the appropriate choice :-

	Column-I		**Column-II**
(A)	Liquid in gas	(i)	Copper dissolved in gold
(B)	solid in solid	(ii)	Glucose dissolved in water
(C)	Liquid in solid	(iii)	Chloroform mixed with nitrogen gas
(D)	Solid in liquid	(iv)	Amalgam of mercury with sodium

(1) (A)→(i), (B)→(iii), (C)→(ii), (D)→(iv)

(2) (A)→(iii), (B)→(i), (C)→(iv), (D)→(ii)

(3) (A)→(ii), (B)→(iii), (C)→(iv), (D)→(i)

(4) (A)→(iii), (B)→(ii), (C)→(i), (D)→(iv)

26) ***Assertion :-*** The van't hoff factor for a weak electrolyte in water depends on its concentration.
Reason :- The degree of dissociation of a weak electrolyte decreases on dilution.

(1) Both Assertion & Reason are correct and Reason is the correct explanation of Assertion.
(2) Both Assertion & Reason are correct but Reason is not the correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

27) **Assertion :–** Isotonic solutions don't show net phenomenon of osmosis. **Reason :–** Isotonic solutions have equal osmotic pressure at constant temperature.

(1) Assertion and reason are true and reason is the correct explanation of assertion.
(2) Assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion and reason are false.

28) Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75 g benzene.

(1) 0.863
(2) 0.556
(3) 4.32
(4) 3.43

29) The molarity of 0.3 N $Ca_3(PO_4)_2$ solution will be:-

(1) 1.8 M
(2) 0.3 M
(3) 0.05 M
(4) 5 M

30) In 2 molal aqueous solution of CH_3OH. Determine mole fraction of H_2O :-

(1) 0.8
(2) 0.7
(3) 0.96
(4) 0.90

31) Find the normality of 15 V H_2O_2 solution :-

(1) 3.42 N
(2) 2.67 N
(3) 4.1 N
(4) 1.98 N

32) V_1 c.c. of solution having molarity M_1 is diluted have molarity M_2. The volume of water required to be added will be :-

(1) $\left(\frac{M_2 - M_1}{M_1}\right) V_1$

(2) $\left(\frac{M_1 - M_2}{M_2}\right) V_1$

(3) $\left(\frac{M_1}{M_2 - M_1}\right) V_1$

(4) $\left(\frac{M_2}{M_2 - M_1}\right) V_1$

33) Equimolal solutions of KCl and compound X in water show depression in freezing point in the ratio of 4 : 1. Assuming KCl to be completely ionized, the compound X in solution must.

(1) dissociate to the extent of 50%

(2) dissociates to the extent of 80%

(3) dimerize to the extent of 50%

(4) trimerize to the extent of 75%

34) Which of the following is not correct option for an ideal solution ?

(1) $\Delta G_{mix} > 0$, $\Delta S_{mix} > 0$

(2) $\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$

(3) $\Delta V_{mix} = 0$, $\Delta S_{mix} > 0$

(4) $\Delta H_{mix} = 0$, $\Delta S_{mix} > 0$

35) If relative decrease in vapour pressure is 0.4 for a solution containing 1 mol AB (electrolyte) in 3 mole water the Degree for dissociation fo AB is

(1) 0.6

(2) 0.8

(3) 0.5

(4) 1

36) Which solution has the highest vapour pressure ?

(1) 0.02 M NaCl at 50°C

(2) 0.03 M sucrose at 15°C

(3) 0.005 M $CaCl_2$ at 50°C

(4) 0.005 M $CaCl_2$ at 25°C

37) If the amount of a non-electrolyte solute is doubled and that of solvent is quadrupled, the elevation in boiling point will be :-

(1) doubled

(2) halved

(3) four times

(4) unchanged

38) For which of the following vant' Hoff's factor is correctly matched -

Salt	Degree of dissociation (a)	i
(1) Na_2SO_4	80 %	2
(2) $K_3[Fe(CN)_6]$	75%	3.25
(3) $[Ag(NH_3)_2]Cl$	50 %	1.8
(4) $[Cr(NH_3)_5Cl]SO_4$	90 %	2.8

(1) 1

(2) 2

(3) 3

(4) 4

39) | $6\%\left(\frac{w}{w}\right)$ of non volatile solute A in water | | $6\%\left(\frac{w}{w}\right)$ of non volatile solute B in water |

If the above solutions are mixed to prepare two solution S_1 and S_2 in the ratio 3 : 2 and 2 : 3 by volume respectively then ratio of $\Delta T_b(s_1)$: $\Delta T_b(s_2)$ is :- Given M_A: M_B = 1 : 3

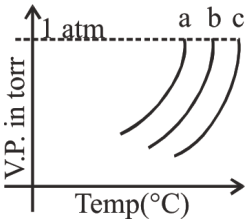
(1) 1 : 1

(2) 4 : 9

(3) 9 : 11

(4) 11 : 9

40) Maximum concentration of solvent present in which solution as shown in the figure. :-



(1) a

(2) b

(3) c

(4) All are some

41) The V.P. of water at 300 K in a closed container is 0.4 atm. If volume of container is halved, its V.P. becomes at 300 K. :-

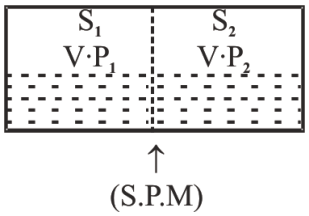
(1) 0.8 atm

(2) 0.2 atm

(3) 0.4 atm

(4) 0.6 atm

42) V.P. of aqueous solution S_1 higher than V.P. of aqueous solution S_2 then :-



(1) More solvent will flow from S_1 to S_2

(2) More solvent will flow from S_2 to S_1

(3) Solvent flow at equal rates

(4) No flow of solent

43) Water should boil at 90°C at pressure :-

(1) 1atm

(2) >1atm

(3) <1 atm

(4) any of these

44) The degree of dissociation (α) of a weak electrolyte A_aB_b is related to van't Hoff factor (i) by the expression :-

(1) $\alpha = \frac{i-1}{a+b+1}$

(2) $\alpha = \frac{i+1}{a+b-1}$

(3) $\alpha = \frac{i-1}{a+b-1}$

(4) $\alpha = \frac{a+b-1}{(i-1)}$

45) Equal masses of two isomers are dissolved in equal mass of a solvent, the solution of 'A' isomer freezer at relatively lower temperature. which isomer 'A or B' has the larger equilibrium constant for ionization. :-

(1) 'A' isomer

(2) 'B' isomer

(3) same for both

(4) unpredicatable

BIOLOGY

1)

Which of the following statement is INCORRECT regarding Porifera ?

(1) Water enters through minute pores (ostia) in the body wall into a central cavity (spongocoel) and goes out through the osculum.

(2) The pathway of water transport is helpful in food gathering, respiratory exchange, and the removal of waste.

(3) Choanocytes or collar cells line the spongocoel only

(4) Digestion is intracellular, and the body is supported by a skeleton made of spongin fibres or spicules.

2)

Read the following statements (A - E) regarding the phylum Cnidaria and select the option that contains all the correct statements.
A. The name cnidaria is derived from the cnidoblasts or cnidocytes (containing stinging capsules or nematocysts) present on the tentacles and body, which are used for anchorage, defense, and capture of prey.
B. They have a central gastro-vascular cavity with a single opening, mouth on hypostome, and their digestion is both extracellular and intracellular.
C. Some of the cnidarians, for example, corals, have a skeleton composed of calcium carbonate.
D. The polyp is a sessile and cylindrical form like Aurelia or jelly fish, whereas medusa is an umbrella-shaped and free-swimming form like Hydra and Adamsia.
E. Those cnidarians which exist in both forms exhibit alternation of generations (Metagenesis), i.e., polyps produce medusae asexually and medusae form the polyps sexually (e.g., Obelia).

(1) A, B, C and E

(2) A, C, D and E

(3) A, B, D and E

(4) B, C, D and E

3) Which one of the following groups of animals is correctly matched with its one characteristic feature without even a single exception :-

(1) Mammalia : give birth to young ones

(2) Reptilia : possess 3-chambered heart with one incompletely divided ventricle

(3) Chordata : possess a mouth provided with an upper and a lower jaw

(4) Chondrichthyes : possess cartilaginous endoskeleton

4) Match **List I** with **List II**.

	List I		**List II**
A.	*Planaria*	I.	Nephridia
B.	*Paramoecium*	II.	Contractile vacuole
C.	*Locusta*	III.	Flame cells
D.	*Hirudinaria*	IV.	Malpighian gland

Choose the **correct** answer from the options give below:

(1) A-I, B-II, C-IV, D-III

(2) A-III, B-II, C-IV, D-I

(3) A-II, B-I, C-IV, D-III

(4) A-I, B-II, C-III, D-IV

5) Read the following statements regarding Ctenophores carefully:
(A) They are exclusively marine organisms, commonly known as sea walnuts or comb jellies.
(B) Their body bears eight internal rows of ciliated comb plates, which help in locomotion.
(C) Digestion in these organisms is both extracellular and intracellular.
(D) Bioluminescence (the property of a living organism to emit light) is well-marked in them.
Choose the correct option regarding the above statements:

(1) Statements (A), (B) and (C) are correct

(2) Statements (A), (C) and (D) are correct

(3) Statements (B), (C) and (D) are correct

(4) All Statements (A), (B), (C) and (D) are correct

6) Given below are two statements regarding Platyhelminthes:
Statement I: Platyhelminthes have a dorso-ventrally flattened body, hence are called flatworms, and these are mostly endoparasites found in animals including human beings.
Statement II: Hooks and suckers are present in the parasitic forms, some of them absorb nutrients from the host directly through their body surface, and specialised cells called flame cells help in osmoregulation and excretion.
Choose the correct option:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

7) Read the following statements regarding Aschelminthes and identify the INCORRECT statement:

(1) The body of the aschelminthes is circular in cross-section, hence they are named roundworms.

(2) They are exclusively parasitic in plants and animals and are never free-living, aquatic, or terrestrial.

(3) The alimentary canal is complete with a well-developed muscular pharynx.

(4) An excretory tube removes body wastes from the body cavity through the excretory pore.

8) How many animals are member of class osteichthyes in given table :

Sawfish, Carcharadon, Lamprey, Betta, Angelfish, Exocoetus, Magur, Trygon, Hagfish, Silverfish, Jellyfish

(1) 6

(2) 4

(3) 5

(4) 7

9) Read the following statements regarding the phylum Annelida and select the **INCORRECT** statement.

(1) Annelids may be aquatic (marine and fresh water) or terrestrial; free-living, and sometimes parasitic, and their body surface is distinctly marked out into segments or metameres (Latin, annulus : little ring).

(2) They possess longitudinal and circular muscles which help in locomotion, and aquatic annelids like Nereis possess lateral appendages, parapodia, which help in swimming.

(3) A closed circulatory system is present in them, and nephridia help in osmoregulation and excretion.

(4) The neural system consists of paired ganglia connected by lateral nerves to a double dorsal nerve cord.

10) Which organisms are first terrestrial chordate as well as vertebrates

(1) Reptiles

(2) Amphibia

(3) Protochordate

(4) Mammal

11) In which one of the following the genus name, its two character and its class are correctly matched :-

	Genus	**Two character**	**Class**
(1)	*Felis*	(a) External ears or pinnae are absent (b) Homeothermic	Mammalia
(2)	*Struthio*	(a) Pleurodont teeth (b) Endoskeleton fully ossified	Aves
(3)	*Bangarus*	(a) epidermal scales (b) Poisonous snake	Reptilia
(4)	*Ichthyophis*	(a) 2 pair of limbs present (b) Poikilothermic animal	Amphibia

(1) 1

(2) 2

(3) 3

(4) 4

12) ***Assertion*** :- Cnidoblasts are present on the tentacles and the body in cnidarians ***Reason*** :- Cnidoblasts are used for anchorage, defence and capture of the prey.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

13) Select the distinguishable characters of amphibians from reptiles :-

(1) Cranial nerves are 10 pairs

(2) Dicondylic skull

(3) Anamniotes animals

(4) All of these

14) Read the following statements regarding the phylum Arthropoda and select the **incorrect** statement.

(1) It is the largest phylum of Animalia which includes insects, and over two-thirds of all named species on earth belong to this phylum.

(2) They have jointed appendages, and their body consists of a head, thorax, and abdomen covered by a chitinous exoskeleton.

(3) Respiratory organs include gills, book gills, book lungs, or a tracheal system, and they possess a closed type of circulatory system.

(4) Sensory organs like antennae, eyes (compound and simple), and statocysts (balancing organs) are present, and excretion takes place through malpighian tubules.

15) Which of the following pairs of animals are similar to each other partaining to feature stated against them?

(1) *Pteropus* and *ornithorhyncus* = viviparity

(2) Garden lizard & crocodile = 4 chamber heart

(3) *Ascaris* & *Ancylostoma* = Metameric segmentation

(4) Sea horse & flying fish = Poikilothermal

16) Which of the following statement is **incorrect**?

(1) Platyhelminthes has incomplete digestive system

(2) In coelenterates, level of organisation is tissue level.

(3) *Nereis* is monoecious but earthworm and leech are dioecious

(4) Simple and compound eyes are present in animals of the phylum which has over two third of all named species on earth.

17) Read the following statements (A to E) regarding the phylum Mollusca and choose the option with all the correct statements:
A. It is the second largest animal phylum and its members are exclusively aquatic (marine or fresh water).
B. Their body is unsegmented, covered by a calcareous shell, and consists of a distinct head, a muscular foot, and a visceral hump.
C. A soft and spongy layer of skin forms a mantle over the muscular foot, and the space between them is called the mantle cavity.
D. Feather-like gills are present in the mantle cavity which have both respiratory and excretory functions.
E. The anterior head region has sensory tentacles, and the mouth contains a file-like rasping organ for feeding, called a radula.

(1) A, C and E

(2) B, D and E

(3) A, B and D

(4) C,D and E

18) Which of the following is not characteristic of chondricthyes ?

(1) Placoid scales
(2) Heterocercal tail
(3) Ventral mouth
(4) 4 pair gills

19) Which is the common between ostrich, penguin and kiwi ?

(1) Running birds
(2) Flightless birds
(3) Hindlimb modify into wings
(4) Migratory birds

20) In which fish locomotion occurs by unpaired fins only ?

(1) *Scolodion*
(2) *Petromyzon*
(3) *Hippocampus*
(4) *Torpedo*

21)

Read the following statements carefully regarding the phylum Echinodermata and select the correct option based on whether they are correct or incorrect:
Statements:
I. They possess an endoskeleton of calcareous ossicles, and hence the name Echinodermata (Spiny bodied).
II. All members of this phylum are marine.
III. Their digestive system is complete, with the mouth located on the upper (dorsal) side and the anus on the lower (ventral) side.
IV. Their most distinctive feature is the presence of a water vascular system, which aids in locomotion, capture and transport of food, and respiration.
V. A specialized excretory system is absent in these animals.

(1) Statements I, II, IV, and V are correct, while Statement III is incorrect.
(2) Statements I, II, III, and V are correct, while Statement IV is incorrect.
(3) Statements I, III, and IV are correct, while Statements II and V are incorrect.
(4) All statements are correct.

22) Digital formula of Hind limb of frog is :

(1) 2 2 3 4 3
(2) 0 2 2 3 3
(3) 2 2 3 3 3

(4) 1 3 3 2 2

23) Teeth in *Rana tigrina* are :-
a. Polyphyodont
b. Acrodont
c. Maxillary and vomerine
d. Homodont

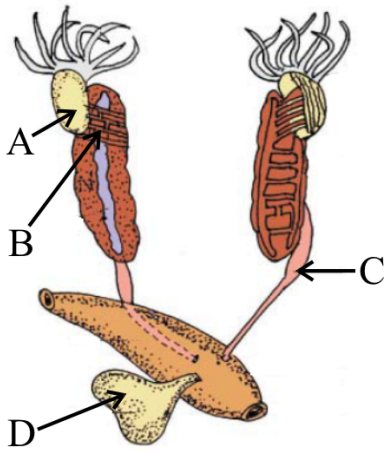
(1) a, c & d

(2) a, b, c & d

(3) a, b & d

(4) b, c & d

24) Identify the A, B, C and D in the given diagram and choose the correct option :



	A	B	C	D
1.	Fat bodies	Vasa deferens	Ureter	Cloaca
2.	Testis	Vasa efferentia	Ureter	Rectum
3.	Testis	Vasa efferentia	Urino genital duct	Urinary bladder
4.	Testis	Vasa deferens	Urino genital duct	Urinary bladder

(1) 1

(2) 2

(3) 3

(4) 4

25) **Assertion :** Hind feet have webbed digits (frogs).
Reason : Hind Limbs help in swimming.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

26) **Assertion :** Frog is tetrapod animal.
Reason : Four limbs are attached to its trunk.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

27) **Statement-1 :** In frog during aestivation and hibernation gaseous exchange takes place through skin.
Statement-2 : In frog gaseous exchange occur only through skin.

(1) Both Statement-I and Statement-II are incorrect.
(2) Both Statement-I and Statement-II are correct.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

28) **Statement-I** The alimentary canal of frog is long.
Statement-II Due to carnivores nature of frog the length of Intestine is reduced.

(1) Both statements are incorrect.
(2) Both statements are correct.
(3) Statement I is correct and statement II is incorrect
(4) Statement I is incorrect and statement II is correct

29) Read the following statements very carefully
(A) In frog digestive food is absorbed by cillia.
(B) Food is captured by the bilobed tongue of frog.
(C) In frog, dissolved oxygen in the water is exchanged through the skin by diffusion.
(D) During Aestivation gaseous exchanges takes place by lungs.
How many of above statements are correct :

(1) One
(2) Two
(3) Three
(4) Four

30) **Assertion (A):** In frog final digestion takes place in the intestine.
Reason (R): Villi and Microvilli found in the inner wall of intestine of frog.

(1) Both Assertion and Reason are true and Reason is NOT the correct explanation of A.
(2) Assertion is true and Reason is false.
(3) Assertion is false and Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of A.

31) Read the following (A-D) statements.
A. It is made of more than one layer (multi-layered) of cells.

B. It has a limited role in secretion and absorption.
C. Their main function is to provide protection against chemical and mechanical stress
D. It covers the dry surface of the skin and the moist surface of buccal cavity.
How many of the above statements are **correct** for compound epithelium :-

(1) Four
(2) Three
(3) Two
(4) One

32) Which of the following statement is/are incorrect ?
(A) Cell junctions are only found in epithelial tissue
(B) Cell junctions provide structural and functional links between neighbouring cells
(C) Adhering junctions perform cementing to facilitate communication between neighbouring cells

(1) Only B
(2) Only A and C
(3) All A, B and C
(4) Only C

33) **Statement A :** Tight junction helps to stop substances from leaking across a tissue.
Statement B : Compound squamous epithelium occurs in stomach, intestine.

(1) Statement A and B, both are correct
(2) Statement A and B, both are incorrect
(3) Statement A is correct and statement B is incorrect
(4) Statement A is incorrect and statement B is correct

34) **Statement-I :** Bones have hard & pliable ground substance.
Statement-II : Cartilage has solid & non-pliable matrix.

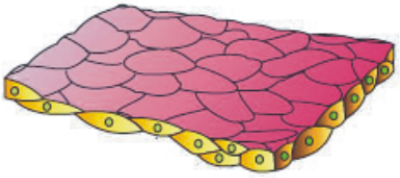
(1) Statement I & II both are correct.
(2) Statement I is correct & II is incorrect
(3) Statement I is incorrect & II is correct
(4) Both statement I & II are incorrect

35) **Statement-I :** Gap Junction facilitates the cells to communicate with each other by connecting the cytoplasm of adjoining cell, for rapid transfer of ions.
Statement-II : Tight Junctions forms cementing layer between the neighbouring cell.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

36) Identify the type of epithelial tissue shown below as well as the related right place of its occurrence in our body along with its correct function and select the correct option for the two

together ?



	Type of epithelial Tissue	Occurance & its function
(1)	Simple squamous Epithelium	- Found in wall of blood vessels and air sacs of lungs. - Function-forming a diffusion boundary
(2)	Simple cuboidal epithelium	- Found in wall of ducts of gland - Function-diffusion and excretion
(3)	Simple columnar epithelium	- Found in wall of nephrones and wall of ducts of glands - Function-secretion and Absorption
(4)	Simple squamous epithelium	- Found in wall of stomach and intestine - Function-absorption and secretion

(1) 1

(2) 2

(3) 3

(4) 4

37) Which of the following excretory structure absent in female cockroach ?

(1) Malpighian tubule

(2) Urecose gland

(3) Fat body

(4) Nephrocyte cell

38) Which of the following is absent in the mouth part of cockroach ?

(1) Labrum

(2) Frons

(3) Labium

(4) Maxillary palps

39) Read the following statements carefully regarding Hemichordates and choose the correct option:

Statement A: Hemichordates have a rudimentary structure in the collar region called a stomochord, which is a structure similar to a notochord.
Statement B: This phylum consists of a small group of worm-like marine animals.
Statement C: Their body is cylindrical and is composed of an anterior proboscis, a collar, and a long trunk.
Statement D: The circulatory system is of the open type and respiration takes place through gills.

(1) All statements (A, B, C, and D) are correct.

(2) Only statements A, B, and C are correct, while D is incorrect.

(3) Only statements A and D are correct, while B and C are incorrect.

(4) All statements (A, B, C, and D) are incorrect.

40) Basic unit in the eye of Cockroach/insect is :-

(1) Retina

(2) Rhabdome

(3) Corneal facet

(4) Ommatidium

41) Which statement is true ?
Statement-I : The entire body is covered by a hard chitinous exoskeleton.
Statement-II : Exoskeleton of each segement consists of chitin plate called sclerites.

(1) Only statement (I) is correct

(2) Only statement (II) is correct

(3) Both statement (I) & (II) are correct

(4) Both statement (I) & (II) are incorrect

42) **Statement-I :** In both male and female cockroach the 10th segment bears a pair of jointed filamentous structure called anal styles.
Statement-II : In females cockroach the 7th sternum is boat shaped.

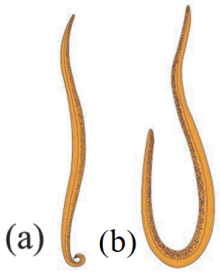
(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

43) Identify the organism in the given diagram and their phylum



	Organism	**Phylum**
(1)	a-male and b-female Ascaris	Aschelminthes

(2)	a-female and b-male Ascaris	Aschelminthes
(3)	a-male earthworm and b-female earthworm	Annelida
(4)	a-female earthworm and b-male earthworm	Annelida

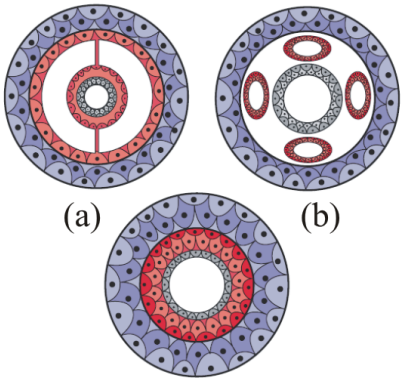
(1) 1

(2) 2

(3) 3

(4) 4

44) Given below are the diagrams and list of examples find out the correct number of examples with



related diagram. (1) Taenia, (2) Ancylostoma, (3) Pheretima, (4) Anopheles, (5) Chaetopleura, (6) Octopus, (7) Cucumeria, (8) Wuchereria, (9) Fasciola.

(1) a-3,4 b-1,2,5 c-6,7,8,9

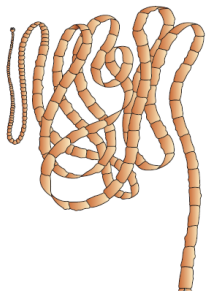
(2) a-1,2 b-3,4,5 c-6,7,8,9

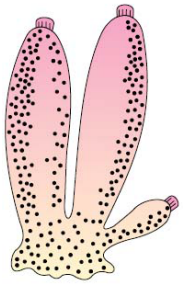
(3) a-3,4,5,6,7 b-2,8 c-1,9

(4) a-5,7 b-2,6,8,9 c-1,3,4

45) Identify the animals given in figure A, B, C and D and select correct option about them ?

(A) 

(B) 

(C)  (D) 

(1) A-*Pila*, B-*Sycon*, C-*Ascaris*, D-Tape worm

(2) A-*Ascaris*, B-Tapeworm, C-*Sycon*, D-*Pila*

(3) A-Tapeworm, B-*Ascaris*, C-*Pila*, D-*Sycon*

(4) A-*Sycon*, B-*Pila*, C-*Ascaris*, D-Tapeworm

ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	3	1	4	2	2	2	3	2	3	2	4	4	3	2	4	4	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	1	1	2	3	1	2	3	3	2	2	4	1	4	3	2	2	4	1
Q.	41	42	43	44	45															
A.	3	1	3	3	1															

BIOLOGY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	4	2	2	1	2	2	4	2	3	1	4	3	4	3	2	4	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	2	3	1	1	3	4	2	1	1	2	3	4	2	1	2	2	1	4
Q.	86	87	88	89	90															
A.	3	3	1	3	2															

CHEMISTRY

1) The vapour pressure of mixture (= 600 mm Hg) is greater than the individual vapour pressure of its constituents (Vapour pressure of CS_2 = 512 mm Hg, acetone = 344 mm Hg). Hence, the solution formed shows positive deviation from Raoult's law.
$\Rightarrow$ (1) $\Delta_{Sol}H > 0$,
(2) Raoult's law is not obeyed
(3) $\Delta_{sol.}$ Volume > 0
(4) CS_2 and Acetone are less attracted to each ether than to themselves.

2) The pure solvent solution will try to maintain higher vapour pressure in the sealed container and in return the solvent vapour molecules will condense in the solution of non-volatile solute as it maintains an equilibrium with lower vapour pressure. (Lowering of vapour pressure is observed when a non volatile solute is mixed in a volatile solvent)
This will eventually lead to increase in the volume of solution and decrease in the volume of solvent.

3) $\frac{\text{Pure milk}}{\text{diluted milk}} = \frac{2}{5}$
If pure milk is 2 cups & diluted milk is 5 cups then water added is 5 – 2 = 3 cups.

4) **Question Explanation:**
The question asks for the mole ratio of benzene and toluene in the vapour phase when the total vapour pressure of the solution is 120 torr.

Concept:
Raoult's Law, Dalton's Law of Partial Pressures, Vapour Phase Composition

Solution:
Assume the solution behaves ideally and contains benzene (Bz) and toluene (Tol).
Given:
$P^0_{benzene} = 150$ torr
$P^0_{toluene} = 50$ torr
Total vapour pressure of solution P = 120 torr
Let x_{Bz} be mole fraction of benzene in liquid, then:
$P = x_{Bz}P^0_{Bz} + (1 - x_{Bz})P^0_{Tol}$
$120 = x_{Bz}(150) + (1 - x_{Bz})(50)$
$120 = 150x_{Bz} + 50 - 50x_{Bz} = 50 + 100x_{Bz}$
$70 = 100x_{Bz} \Rightarrow x_{Bz} = 0.7, \quad x_{Tol} = 0.3$
Now vapour-phase mole fractions:
$y_{Bz} = \frac{x_{Bz}P^0_{Bz}}{P} = \frac{0.7 \times 150}{120} = \frac{105}{120} = 0.875$
$y_{Tol} = 1 - 0.875 = 0.125$
So the mole ratio of benzene : toluene in vapour phase is

$$\frac{y_{Bz}}{y_{Tol}} = \frac{0.875}{0.125} = 7:1$$

Final Answer: Option(1)

5)

[I] 0.1 m KCl [strong electrolyte]
α = 1 $KCl \rightarrow K^+ + Cl^-$
n = 2
⇒ i = 1 + α (n–1) = 1 + 1 (2–1)
⇒ i = 2
⇒ $(\Delta T_f)_I = i \times m \times k_f = 2 \times 0.1\ k_f = 0.2\ k_f \Rightarrow (T_f)_I = -0.2\ k_f$
(II) 0.1 m HA (a = 0.5)
$HA \rightleftharpoons H^+ + A^-$
n=2
⇒ i = 1 + α (n–1)
⇒ i = 1+ 0.5 (2–1)
⇒ i = 1.5
⇒ $(\Delta T_f)_{II} = imk_f = 1.5 \times 0.2\ k_f = 0.3\ k_f \Rightarrow (T_f)_{II} = -0.3\ k_f$
(III) 0.1 m $C_6H_{12}O_6$ (not dissociated)
⇒ i = 1
⇒ $(\Delta T_f)_{III} = imk_f = 1 \times 0.1\ k_f = 0.1\ k_f \Rightarrow (T_f)_{III} = -0.1k_f$
⇒ $(T_f)_{II} < (T_f)_I < (T_f)_{III}$
⇒ option [D]

6)

As $P_A^o > P_B^o$
A is more volatile hence more vapour of a there as compared to liquid A.

7)

Ans : → Given : → moles of sucrose = x moles
mass of water (solvent) = 100 g
molality = $\frac{x}{100} \times 1000$ = 10 x mol/kg
Freezing point = –0.2°C
Depression in freezing point. = ΔT_f = 0.2°C
$\Delta T_f = K_f\ m$
$K_f = \frac{\Delta T_f}{m} = \frac{0.2}{10x} = \frac{0.02}{x}$
⇒ New temperature = –0.25°C
$\Delta T_r' = 0.25°C$
$K_F = \frac{0.02}{x}$, mass of solvent left = Wg
moles of sucrose = x moles
$\Delta T_f' = 0.25 = K_f \times m$

$$0.25 = \frac{0.02}{x} \times \frac{x \times 1000}{w}$$

$$W = \frac{0.02 \times 1000}{0.25} = 80\ g$$

Mass of ice separated = initial mass of solvent – mass of solvent left

= 100 – 80 = 20g – B.

8) **Explain the Question:** Calculate the degree of dissociation α of an electrolyte using osmotic pressure.

Concept: Osmotic pressure with the Van't Hoff factor: $\pi = iCRT$, where $i = 1 + (n-1)\alpha$.

Solution:

For the electrolyte A_2B, the dissociation reaction is $A_2B \rightarrow 2A^+ + B^{2-}$, which gives n = 3 particles. The Van't Hoff factor is $i = 1 + (3-1)\alpha = 1 + 2\alpha$.

Using the osmotic pressure formula $\pi = iCRT$, we convert the temperature to Kelvin: T = 27 + 273 = 300K.

Substituting the given values:

$$6.15 = (1 + 2\alpha) \times 0.1 \times 0.082 \times 300$$

$$6.15 = (1 + 2\alpha) \times 0.1 \times 24.6$$

$$6.15 = (1 + 2\alpha) \times 2.46$$

$$1 + 2\alpha = \frac{6.15}{2.46} = 2.5$$

$$2\alpha = 1.5 \Rightarrow \alpha = 0.75$$

Final Answer: Option (2)

9) $$\pi = \frac{mass}{v_{cc}} \times \frac{1000}{mm}RT$$

$$\pi = c \times \frac{1000}{mm}RT$$

$$m = \frac{1000}{mm}RT$$

$mm = 4.9 \times 10^6$

10) $$\alpha = \frac{i-1}{\frac{1}{2}-1} \Rightarrow \frac{0.52-1}{0.5-1} = 0.96$$

π = iCRT

= 0.52 × 0.1 × 0.0821 × 300

= 1.28 atm

11) **Question Explanation:**

The question asks to determine the vapor pressures of pure components $A\ (P_A^\circ)$ and $B\ (P_B^\circ)$ using the given total vapor pressure equation of a binary liquid mixture in terms of the mole fraction of component $B\ (X_B)$.

Concept:

Raoult's Law, Vapor Pressure of Liquid Solutions

Solution:

The given relation for total vapor pressure is:

$P_{Total} = 120 - 75X_B$

1. For pure component A:

The mole fraction of component B is zero ($X_B = 0$).

$P_A^\circ - 120 - 75(0) - 120$ Torr

2. For pure component B:

The mole fraction of component B is one ($X_B = 1$).

$P_B^\circ - 120 - 75(1) - 45$ Torr

Thus, the vapor pressures of pure A and B are 120 Torr and 45 Torr, respectively.

Final Answer: Option (3)

12) **Question Explanation:**

The question asks to identify which pair of liquids among the given options does not form an ideal solution.

Concept:

An ideal solution is formed when solute and solvent molecules have nearly identical molecular sizes, structures, and intermolecular forces $(A - A \approx B - B \approx A - B)$.

Solution:

1. C_6H_6 and $C_6H_5CH_3$ (Benzene + Toluene):

Both are non-polar aromatic hydrocarbons with nearly identical structures and intermolecular forces. They form an ideal solution.

2. C_2H_5Cl and C_6H_5OH (Ethyl chloride + Phenol):

Phenol forms strong intermolecular hydrogen bonds, whereas ethyl chloride only exhibits dipole-dipole interactions. Mixing them significantly alters the intermolecular forces, so they do not form an ideal solution (forms a non-ideal solution).

3. C_6H_5Cl and C_6H_5Br (Chlorobenzene + Bromobenzene):

Both are haloarenes with similar molecular structure and polarity. They form an ideal solution.

4. C_2H_5Br and C_2H_5I (Ethyl bromide + Ethyl iodide):

Both are alkyl halides with nearly identical structures and interactions. They form an ideal solution.

Final Answer: Option (2)

13) Isotonic solution has same conc.

$\pi_1 = \pi_2 \quad C_1 = C_2 \quad n_1 = n_1 \quad \frac{W_1}{M_1} = \frac{W_2}{M_2}$

So $\frac{x}{180} = \frac{4}{60} \Rightarrow$ x = 12 g Ans.

(b) For isotonic solution $\pi_1 = \pi_2$; $C_1 = C_2$; $n_1 = n_2$

$\frac{W_1}{M_1} = \frac{W_2}{M_2} \Rightarrow \frac{1.05}{M} = \frac{30}{180}$ M = $\frac{1.05 \times 30}{180}$ = 63 Ans.

14) All solution have same No. of particle and also have same value of π. $n_1 = n_2$; $\pi_1 = \pi_2$ (Isotonic)

15) **Question Explanation:** The question asks which relation between liquid-phase and vapour-phase mole fractions is correct for an ideal solution of two volatile liquids.

Concept : Raoult's Law, Dalton's Law of Partial Pressures, Vapour Phase Composition
Solution:
For an ideal solution,
$y_M = \frac{x_M P_M^o}{P_{total}}$, $y_N = \frac{x_N P_N^o}{P_{total}}$
So,
$\frac{y_M}{y_N} = \frac{x_M P_M^o}{x_N P_N^o}$
Given,
$P_M^o = 450\,mmHg,\ P_N^o = 700\,mmHg$
Hence,
$\frac{y_M}{y_N} = \frac{x_M}{x_N}.\frac{450}{700}$
Since
$\frac{450}{700} < 1$
therefore,
$\frac{y_M}{y_N} < \frac{x_M}{x_N}$
So the correct statement is
$\frac{x_M}{x_N} > \frac{y_M}{y_N}$
Final Answer : Option (3)

16) **Question Explanation:**
Find degree of dissociation of silver nitrate.

Concept:
This question is based on Van't Hoff factor.

Solution:
$M_{avg} = \frac{M_{calculated}}{i}$
$M_{avg} = \frac{M_{calculated}}{1 + (n-1)\alpha}$
⇒ for $AgNO_3(aq) \rightarrow Ag^+_{(aq.)} + NO^-_{3(aq.)}$
n = 2 ;
⇒ i = 1 + (2 – 1)α = (1 + α)
M_{avg} = 92.64; $M_{Calculated}$ = 170
⇒ Substituting in the equation
$92.64 = \frac{170}{1 + \alpha} \Rightarrow \alpha = 0.835$
%α = 83.5%

Final Answer: The correct option is (2).

17) As per Henry's law
$x_{Ar} = \frac{P_{Ar}}{K_H} = \frac{0.403}{40.3 \times 10^3} = 1 \times 10^{-5}$

$\Rightarrow \frac{n_{Ar}}{n_{Ar}+n_{H_2O}} = 1\times10^{-5}$ [1 litre of H_2O= 55.56 mol]

$\Rightarrow \frac{n_{Ar}}{55.56} = 1\times10^{-5}$

$\Rightarrow n_{Ar} = 55.56\times10^{-5}$ mol

18) V_{H_2O} = 1L

$\Rightarrow W_{H_2O}$= 1kg [$\because d_{H_2O}$ = 1 g/mL]

$\Rightarrow \Delta T_f = m.kf$

$\Rightarrow \frac{\Delta T_f}{k_f} = m = \frac{1}{1000}$

$\Rightarrow m = \frac{n_{glucose}}{W_{H_2O(kg)}} = \frac{n}{1} = \frac{1}{1000}$

$\Rightarrow n = \frac{1}{1000}$ [$\because$ M_w of glucose = 180 g/mol]

$\Rightarrow W_{glucose} = nM_w = \frac{1}{1000}\times 180 = 0.18$ g

$\Rightarrow W_{glucose}$ = 0.18 g

$\Rightarrow$ option [D]

19) **Question Explanation:** Distillation of an azeotrope.

Concept: Azeotropes

Solution: An **azeotrope** is a special type of solution that boils at a constant temperature and distills without any change in its composition. The given mixture of water and nitric acid is a **maximum boiling azeotrope**, meaning its boiling point (393.5 K) is higher than that of its individual components (373 K and 359 K). When distilled, the liquid and vapor phases have the same composition, so the mixture distills over as a whole.

Final Answer: both of them will distill over in the same composition as they are in the mixture.

20) **Question Explanation :** Find correct plot for the given data.
Concept : Henrys law
Solution:
According to Henry's law;
$P_i = K_H . X_i$; where
P_i = partial pressure of gas above the solution
X_i = mole fraction of gas in the solution
K_H = Henry's constant
$\Rightarrow P_i = K_H (1 - X_{H_2O})$
$\Rightarrow P_i = (-K_H) X_{H_2O} + K_H$; comparing with
Y = (m)X + C ;
slope = $-K_H$; y-interecpt = K_H
Final Answer : Option (3)

21) **Question Explanation :** Find statement which is incorrect
Concept : Non ideal solution
Solution:
1. Statement 1 is incorrect. While standard air contains N_2 and O_2, the tanks specifically used by scuba divers are filled with air diluted with helium (11.7% He, 56.2 % N_2, and 32.1% O_2) to avoid decompression sickness known as "the bends" and the toxic effects of high nitrogen concentration in the blood.
2. Statement 2 is correct. Edema is the medical term for swelling caused by excess fluid trapped in your body's tissues, often due to high salt intake leading to water retention.
3. Statement 3 is correct. Osmotic pressure measurements are carried out at room temperature, making them ideal for determining the molecular mass of biomolecules (like proteins) that are unstable at high temperatures.
4. Statement 4 is correct. A mixture of 68% HNO_3 and 32% H_2O by mass forms a maximum boiling azeotrope with a boiling point of 393.5 K .
Final Answer : Option (1)

22) **For A :** 100 gm solution → 2 gm solute A

$\therefore$ Molality = $\frac{2/M_A}{0.098}$

For B : 100 gm solution → 8 gm solute B

$\therefore$ Molality = $\frac{8/M_B}{0.092}$

$\because (\Delta T_B)_A = (\Delta T_B)_B$

$\therefore$ Molality of A = Molality of B

$$\frac{2}{0.098M_A} = \frac{8}{0.092M_B}$$

$$\frac{2}{98} \times \frac{92}{8} = \frac{M_A}{M_B}$$

$$\frac{1}{4.261} = \frac{M_A}{M_B}$$

$$\therefore M_B = 4.261 \times M_A$$

23)

Conceptual

24)

	value of 'i'
Urea	1
NaCl	2
Na_2SO_4	3
K_2SO_4	3
Sugar	1
Glucose	1
KCl	2

25)

Question Explanation: Match each type of solution (solute–solvent) with its correct real-life example.

Concept: Classification of solutions based on physical state of solute and solvent.

Solution:

(A) **Liquid in gas** → Chloroform (liquid) mixed with nitrogen gas → **(iii)**

(B) **Solid in solid** → Copper (solid) dissolved in gold (solid) → **(i)**

(C) **Liquid in solid** → Amalgam of mercury (liquid) with sodium (solid) → **(iv)**

(D) **Solid in liquid** → Glucose (solid) dissolved in water (liquid) → **(ii)**

Correct match - (A)→(iii), (B)→(i), (C)→(iv), (D)→(ii)

Final Answer: Option (2)

26)

A-i depends on α for weak electrolyte and α depends on concentration of electrolyte R-on dilution α of weak electrolyte increases.

27) **Question Explanation:**
Determine if the given assertion and reason regarding isotonic solutions and osmosis are true, and if the reason correctly explains the assertion.

Concept:
Isotonic solutions are defined as solutions having equal osmotic pressure ($\pi_1 = \pi_2$) at the same temperature. Because the chemical potential of solvent is equal on both sides, there is no driving force for solvent movement, resulting in no net osmosis.

Solution:
1. **Assertion: True.** When two isotonic solutions are separated by a semi-permeable membrane, solvent molecules pass in both directions at equal rates, resulting in no net movement of solvent (no net osmosis).
2. **Reason: True.** By definition, isotonic solutions have equal osmotic pressure ($\pi_1 = \pi_2$) at a constant temperature.
3. **Explanation Check:** Since the equality of osmotic pressures ($\pi_1 = \pi_2$) prevents any pressure or concentration gradient across the membrane, it directly explains why no net osmosis occurs.

Final Answer: Option (1)

28)

Question Explanation: Calculation of the Molality of the solution.
Concept : This question is based on Molality.

Solution :
Molality is defined as the number of moles of solute per kilogram of solvent.
Moles of ethanoic acid = mass/molar mass=2.5 g/(60 g/mol) =0.04167 mol
Molality = moles of solute /mass of solvent (in kg) =0.04167 mol/0.075 kg =0.5556 mol/kg
Final Answer : option (2)

29) **Question Explanation :** We need to find molarity of $Ca_3(PO_4)_2$ solution using given Normality.
Concept : Relation between Normality and molarity.
Solution :
 $N = M \times$ n-factor
$Ca_3(PO_4)_2 \Rightarrow$ n-factor = + 6

$$M = \frac{N}{n-factor} = \frac{0.3}{6} = 0.05 \text{ M}$$

Final Answer : Option (3)

30) Here,
2 mole of CH_3OH is dissolved in 1 kg of H_2O

$$n_{H_2O} = \frac{1000}{18} = 55.5 \text{ mol}$$

Total mol = 55.5 + 2 = 57.5

$$X_{H_2O} = \frac{n_{H_2O}}{n_{total}} = \frac{55.5}{57.5} = 0.96$$

31)

Explain Question – Find the normality of a 15 V H_2O_2 solution.
Concept – This question based on Volume strength of H_2O_2.
Solution –
Normality (N) = (Volume Strength) / 5.6
Normality (N) = 15 V / 5.6
Normality (N) ≈ 2.67857 N or ≈ 2.67 N
Final Answer – Option (2)

32) **Explain question:**
To calculate the volume of water required to achieve new molarity M_2.
Concept:
Dilution and mixing of solutions.
Solution:
By dilution moles of solute do not change, only molarity changes.
Let V_2 c.c. water id added to the solution, then, initial moles = final moles.
$V_1 M_1 = (V_1 + V_2) M_2$
$\Rightarrow V_1 M_1 = V_1 M_2 + V_2 M_2$
$\Rightarrow V_2 M_2 = V_1 (M_1 - M_2)$

$$V_2 = \frac{V_1}{M_2} (M_1 - M_2)$$

$V_2 = \left(\frac{M_1 - M_2}{M_2}\right) V_1$

Final answer: option (2)

33) $\frac{(\Delta T_f)_{KCl}}{(\Delta T_f)_x} = \frac{2 \times k_f \times 1}{i \times k_f \times 1}$

$\frac{4}{1} = \frac{2}{(i)_x} \Rightarrow \boxed{(i)_x = 0.5}$

$i = 1 - \alpha + \frac{\alpha}{n}$ if n = 3

$0.5 = 1 - \alpha + \frac{\alpha}{3} \Rightarrow \alpha = 0.75$

34)

Question Explanation: Asking about incorrect condition for an ideal solution.
Concept: Ideal solution
Solution:
For an ideal solution:
• $\Delta H_{mix} = 0$
• $\Delta V_{mix} = 0$
• $\Delta S_{mix} > 0$
• $\Delta G_{mix} = \Delta H_{mix} - T\Delta S_{mix}$
Since $\Delta H_{mix} = 0$ and $\Delta S_{mix} > 0$
$\Rightarrow \Delta G_{mix} = -T\Delta S_{mix} \Rightarrow \Delta G_{mix} < 0$
Option (1): $\Delta G_{mix} > 0$, $\Delta S_{mix} > 0$ (Wrong because ΔG_{mix} must be negative)
Option (2): $\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$ (Correct)
Option (3): $\Delta V_{mix} = 0$, $\Delta S_{mix} > 0$ (Correct)
Option (4): $\Delta H_{mix} = 0$, $\Delta S_{mix} > 0$ (Correct)
Final Answer: (1)

35) $\frac{\Delta P}{P^0} = \frac{in_B}{in_B + n_A}$ $n_B=1$, $n_A=3$

$0.4 = \frac{in}{in+3} = \frac{i}{i+3} = 0.4$ $\boxed{i = 2}$

$AB \rightleftharpoons A^+ + B^-$ $i = 1 + \alpha = 1 - \alpha$ α α

$\boxed{\alpha = 1}$

36) For high vapour pressure concentration of solute should be low and temperature should be high.

37) $\Delta T_b \propto$ molality

$\propto \left(\frac{\text{Mass of solute/Mol. mass of solute}}{\text{Mass of solvent}}\right)$

 If solute = doubled
solvent = quadrupled

then, ΔT_b = Halved

38)

$K_3[Fe(CN)_6] \rightarrow 3K + [Fe(CN)_6]^{-3}$

1–α 3α α

$$i = \frac{1-\alpha+3\alpha+\alpha}{1} = (1+3\alpha)$$

$i = 1 + 3(0.75) \Rightarrow 3.25$

39) $\frac{\Delta T_b(S_1)}{\Delta T_b(S_2)} = \frac{K_b m_1}{K_b m_2} = \frac{m_1}{m_2} \Rightarrow$

Molality for S_1 solution

$$m_1 \rightarrow S_1 \Rightarrow \frac{\frac{3w}{M_A}+\frac{2w}{M_B}}{5} =$$

$$m_2 \rightarrow S_2 \Rightarrow \frac{\frac{2w}{M_A}+\frac{3w}{M_B}}{5}$$

$$\frac{m_1}{m_2} = \frac{\frac{3w}{M_A}+\frac{2w}{M_B}\Big/5}{\frac{2w}{M_A}+\frac{2w}{M_B}\Big/5} = \frac{\frac{3wM_B+2wM_A}{5M_AM_B}}{\frac{2wM_B+3wM_A}{5M_AM_B}}$$

$$= \frac{3M_B+2M_A}{2M_B+3M_A} = \frac{3+2\left(\frac{M_A}{M_B}\right)}{2+3\left(\frac{M_A}{M_B}\right)}$$

$$\frac{3+2\times\frac{1}{3}}{2+3\times\frac{1}{3}} = \frac{11}{3\times 3} = \frac{11}{9}$$

40) Concentration of solute c > b > a
Concentration of solvent a > b > c

41) V.P. depends only on temperature.

42) Concentration ($S_2 > S_1$) therefore solvent moves lower conc. to Higher conc.

43) Fact

44) $A_aB_b \Rightarrow aA^{+b} + bB^{-a}$

1 – α aα bα

$$i = \frac{1-\alpha+a\alpha+b\alpha}{1}$$

$$\frac{i-1}{a+b-1}=\alpha$$

45) $(\Delta T_f)_A > (\Delta T_f)_B$ i.e D.O.D (∝) is higher for 'A' therefore K_{eq} will Higher.

BIOLOGY

46)

NCERT, Pg. # 40

47)

NCERT, Pg. # 41

48)

NCERT, Pg. # 47

49)

NCERT, Pg. # 42, 43, 44

50)

NCERT, Pg. # 42

51)

NCERT, Pg. # 42

52)

NCERT, Pg. # 43

53)

NCERT XI, Pg. No. 57

54)

NCERT, Pg. # 43

55) NCERT-XII, Pg. # 57

56) NCERT–XI, Pg. # 58

57)

NCERT, Pg. # 41

58)

NCERT, Pg. # 48, 49

59)

NCERT, Pg. # 44

60)

NCERT XI Pg.# 48

61)

NCERT, Pg. # 41, 42, 43, 44

62)

NCERT, Pg. # 44

63)

NCERT, Pg. # 47, 48

64)

NCERT, Pg. # 50

65)

NCERT, Pg. # 47

66)

NCERT, Pg. # 45

67)

Pg. # 190 Module # 4

68)

Pg. # 189, Module # 4

69)

NCERT, Pg. # 119

70)

NCERT, Pg. # 116

71)

NCERT, Pg. # 116

72)

NCERT, Pg. # 116

73) NCERT Pg. No. # 116

74) NCERT Pg. No. # 117, 118

75)

NCERT-XI, Pg. # 117

76)

NCERT, Pg. # 102

77) **Question Explanation**: The question asks to identify incorrect statements about cell junctions and their functions.

Concept: The question is based on intercellular junctions.

Solution: The incorrect statements are A and C.

A. (A) Cell junctions are only found in epithelial tissue → Incorrect. Cell junctions are common in epithelial tissue, but also occur in other tissues like cardiac muscle and nervous tissue.
B. (B) Cell junctions provide structural and functional links between individual cells → Correct. They help in adhesion, communication, and barrier formation.
C. (C) Adhering junctions perform cementing to facilitate communication between neighbouring cells → Incorrect. Adhering junctions provide mechanical strength, not communication. Gap junctions facilitate intercellular communication.

Final Answer: Option 2

78)

Question Explanation: Check which statement is true and which is false.

Concept: Epithelium

Solution:

Statement A = Correct – Tight junctions prevent leakage between cells.

Statement B = Incorrect – Compound (stratified) squamous epithelium is found in skin, not in stomach, intestine, or trachea.

Final answer: Option 3

79) NCERT Pg. # 104

80) MODULE Pg No : 67

81) **Question Explanation**: You must identify which epithelial tissue is shown and match it with its correct location and function.
Concept: Types of epithelium
Solution: Simple squamous epithelium is thin and flat → found in blood vessels, air sacs, and helps in diffusion. So,Option 1 correctly describes simple squamous epithelium, its location, and its function.

Final answer: Option 1

82) NCERT Pg.#114 (E), 114 (H)

83)

Frons

84)

NCERT, Pg. # 45

85)

NCERT XI, Pg. # 114

86)

NCERT, Pg. # 111

87) NCERT-XI Pg. # 112

88)

NCERT, Pg. # 43

89)

NCERT, Pg. # 39

90) NCERT Pg.# 52,51,49,53 Figure- 4.10(b), 4.9(a), 4.5(a), 4.13(a)